Quantum, Not-So-Computing: A Variational Quantum Classifier on Moons

Write-up for github.com/supernova-sys/Quantum-not-so-computing

1. Problem

The task is binary classification of the two-class *make_moons* dataset, a small non-linearly separable toy problem. The question the repo asks is deliberately modest: can a variational quantum classifier (VQC) built in Qiskit match classical models that have a **similar number of trainable parameters**? It is not an attempt to show quantum advantage.

Data. 100 points, noise 0.2, seed 42, with a stratified 70/30 train/test split (30 test points). Features are scaled to $[0, \pi]$ so they can be used directly as rotation angles. Accuracy is the only metric reported.

2. Method

Six models are compared, with parameter counts matched as closely as practical.

- **Classical baselines:** logistic regression (3 parameters); logistic regression with x^2 terms (6); a small neural net with 2 tanh hidden units (9); and an RBF SVM (29), kept as a reference ceiling rather than a parameter-matched rival.
- **Ready-made VQC:** Qiskit's *ZZFeatureMap* with a *RealAmplitudes* ansatz (8 parameters), trained with COBYLA for 100 iterations.
- **From-scratch VQC:** RY feature encoding, 4 ansatz layers with CNOT entanglers (8 parameters). The prediction is the probability of measuring 1 on qubit 0, trained with binary cross-entropy and COBYLA for 200 iterations. Circuits are simulated exactly with *Statevector*.
- **Robustness check:** every model is re-run over 10 different train/test splits, so conclusions do not hinge on one lucky split.

Seeds are fixed (SEED = 42 plus the split index). Tested versions: qiskit 2.6.0, qiskit-machine-learning 0.9.1, scikit-learn 1.6.1, numpy 2.1.3, scipy 1.16.3, matplotlib 3.10.0. All code is in *vqc_moons_notebook.ipynb*.

3. Results

Single split (seed 42), test accuracy:

Model	Params	Test acc.
Logistic regression	3	0.83
Logistic regression + x^2	6	0.83
Small neural net (2 tanh)	9	0.73
RBF SVM (reference)	29	0.93
VQC, ready-made (ZZ)	8	0.60
VQC, from scratch (RY)	8	0.83

Mean test accuracy over 10 splits:

Model	Mean acc.	Std
RBF SVM (reference)	0.913	0.034
Logistic regression	0.850	0.056
Small neural net	0.850	0.052
VQC, from scratch (RY)	0.840	0.061
Logistic regression + x^2	0.840	0.061
VQC, ready-made (ZZ)	0.567	0.083

Reading the numbers. The from-scratch VQC (0.840) is statistically indistinguishable from the small classical models (0.840 to 0.850) and does not beat them. The RBF SVM is clearly the strongest model. The ready-made ZZ-based VQC performs close to chance (0.567), while the simpler RY encoding works far better with the same parameter count. The notebook also includes a decision-boundary plot comparing the from-scratch VQC with the SVM.

4. Honest limits

- **Tiny test set.** With 30 test points, split-to-split standard deviation is about 0.05 to 0.06. Differences of a few percentage points among the small models are within noise, so no ranking among them should be claimed.
- **Ideal simulation only.** Everything uses exact statevector simulation: no shot noise and no hardware noise. Results say nothing about how the classifier would behave on a real device.
- **The ZZ failure is under-explained.** The README attributes it to the short COBYLA run (100 iterations) and the 2-qubit circuit, but this was not isolated with an ablation. The result applies to this particular circuit and setup, so it should not be read as a verdict on ZZFeatureMap or on quantum models in general.
- **Unequal training budgets.** The two VQCs received different COBYLA iteration budgets (100 vs 200), so the ZZ-vs-RY gap mixes circuit design with optimization effort.
- **Parameter count is a loose proxy.** Matching trainable parameters does not match expressivity, training difficulty or compute cost. The SVM's 29 parameters are not comparable in kind to the others.
- **Single toy dataset, accuracy only.** One 2-D problem, 100 samples, no calibration, precision/recall or other datasets. Nothing here supports claims of quantum advantage.

Bottom line: on this toy problem a small, well-chosen VQC is competitive with similarly sized classical models but offers no measurable gain.